Practice Quiz: Agents (Mathematical Frameworks)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. Exact Bayesian inference is intractable because: A) Bayes' theorem is incorrect B) Computing P(o) requires summing over every possible state, which grows exponentially C) Probabilities can't be computed D) The brain doesn't use probability

2. The recognition model q(s) is: A) The exact posterior distribution B) An approximate posterior — the agent's best guess about hidden states C) The prior distribution D) The likelihood function

3. KL divergence D_KL[q(s) || P(s | o)] equals zero when: A) q is very different from P B) q(s) perfectly matches the true posterior P(s | o) C) The agent has no data D) All states are equally likely

4. The ELBO decomposes into: A) Prior + Posterior B) Accuracy (how well q explains data) minus Complexity (how far q is from the prior) C) Reward minus Cost D) Input minus Output

5. Variational free energy F is: A) Physical energy in the brain B) The negative ELBO — minimizing F is equivalent to good inference C) Always zero D) The temperature of the system

6. The accuracy-complexity trade-off means: A) Always prefer the most complex model B) Balance fitting the data well with keeping the model simple C) Accuracy and complexity are the same thing D) Ignore complexity entirely

7. Variational inference turns inference into: A) A search problem B) An optimization problem — adjusting q(s) to minimize free energy C) A random guess D) Exact computation

Part B: Short Answer

1. Explain in your own words why the brain must use approximate inference rather than exact Bayesian inference. Use a concrete example to illustrate the computational challenge.

2. Consider two approximate posteriors for the state "is it raining?":

- q_1: P(rain) = 0.7, P(no rain) = 0.3
- q_2: P(rain) = 0.9, P(no rain) = 0.1

If the true posterior is P(rain | data) = 0.8, which approximation has lower KL divergence? Why?

3. Explain the ELBO accuracy-complexity trade-off using a real-world analogy. For example, compare a weather forecast model that uses 3 variables vs. one that uses 100 variables.